Dataset Head:

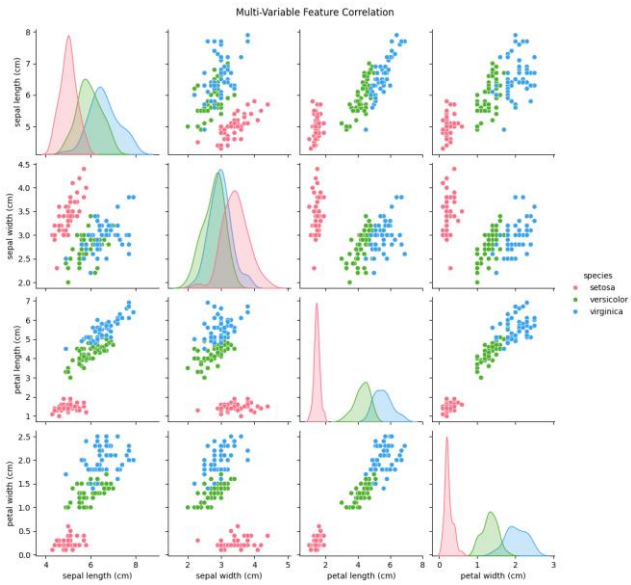
| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) \ |
|---|---|---|---|---|
| 0 | 5.1 | 3.5 | 1.4 | 0.2 |
| 1 | 4.9 | 3.0 | 1.4 | 0.2 |
| 2 | 4.7 | 3.2 | 1.3 | 0.2 |
| 3 | 4.6 | 3.1 | 1.5 | 0.2 |
| 4 | 5.0 | 3.6 | 1.4 | 0.2 |

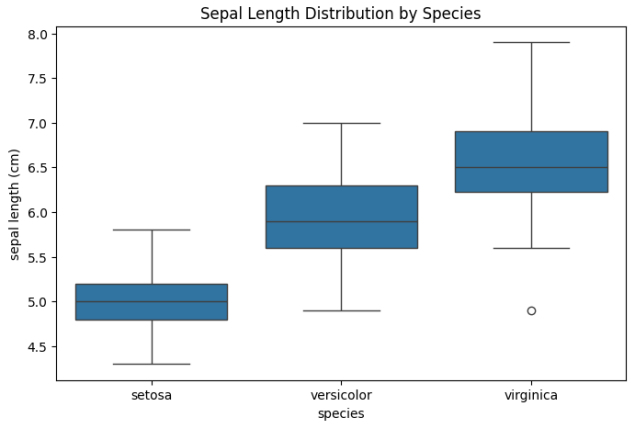
| | species |
|---|---|
| 0 | setosa |
| 1 | setosa |
| 2 | setosa |
| 3 | setosa |
| 4 | setosa |

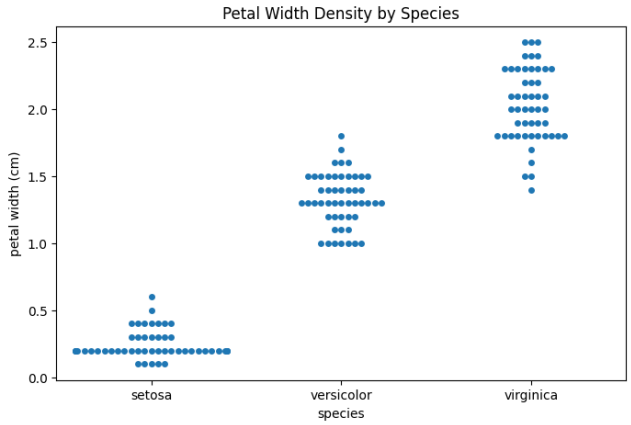
--- Statistical Significance Test ---

T-Statistic: -10.5210, P-Value: 0.0000

Conclusion: The difference in sepal length is STATISTICALLY SIGNIFICANT.



/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarning: 12.0% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.

warnings.warn(msg, UserWarning)